

MATHEMATICAL INDUCTION



SYNOPSIS

➤ Principle of Finite Mathematical

Induction:

For $n \in \mathbb{N}$, let $P(n)$ be a statement in terms of n . If $P(1)$ is true and $P(k)$ is true $\Rightarrow P(k+1)$ is true, then $P(n)$ is true, for all $n \in \mathbb{N}$.

➤ Principle of Complete Mathematical Induction:

For $n \in \mathbb{N}$, let $P(n)$ be a statement in terms of n . If $P(1), P(2), P(3), \dots, P(k-1)$ are true $\Rightarrow P(k)$ is true, then $P(n)$ is true, for all $n \in \mathbb{N}$.

➤ $a, (a+d), (a+2d), \dots$ form an A.P. then

(i) n^{th} term $t_n = a + (n-1)d$, Where a is the first term and d is the common difference.

(ii) Sum of n terms $S_n = \frac{n}{2} [2a + (n-1)d]$

$$= \frac{n}{2} [a + l]$$

Where a = first term, l = last term

➤ $a, ar, ar^2, \dots$ form a G.P then

(i) n^{th} term $t_n = ar^{n-1}$, Where a = first term
 r = common ratio

(ii) Sum of n terms $S_n = a \frac{(r^n - 1)}{r - 1}$;

(iii) In an infinite G.P, Sum of Infinite terms is

$$S_\infty = \frac{a}{1-r} \text{ where } |r| < 1$$

➤ $a + (a+d)r + (a+2d)r^2 + \dots$ $\dots + [a + (n-1)d]r^{n-1}$ form A.G.P. then

(i) n^{th} term $t_n = [a + (n-1)d]r^{n-1}$

(ii) Sum of n terms

$$S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{(1-r)}$$

(iii) Sum of Infinite terms

$$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2} \text{ where } |r| < 1$$

W.E-1: Sum to infinite terms of

$1 + 3x + 5x^2 + 7x^3 + \dots + \infty$ (If $|x| < 1$) is ____
[Eam-1998]

Sol: $S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

where $a = 1, d = 2, r = x$

$$= \frac{1}{1-x} + \frac{2x}{(1-x)^2} = \frac{1+x}{(1-x)^2}$$

Similarly If $|x| < 1$,

$$S_\infty = 1 + 2x + 3x^2 + 4x^3 + \dots + \infty = \frac{1}{(1-x)^2}$$

SOME IMPORTANT POINTS

i) Sum of first n natural numbers i.e.

$$\sum n = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}, \forall n \in \mathbb{N}$$

ii) Sum of the squares of first n natural numbers is

$$\sum n^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}, \forall n \in \mathbb{N}$$

iii) Sum of the cubes of first n natural numbers is

$$\sum n^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$$

$$= \left(\sum n\right)^2, \forall n \in \mathbb{N}$$

iv) $1^4 + 2^4 + 3^4 + \dots + n^4 = \sum n^4$

$$= \frac{n(n^4 - 1)}{5} + \frac{n(n+1)(3n^2 - n + 1)}{6}$$

v) Sum of the first ' n ' odd +ve integers =

$$1 + 3 + 5 + \dots + (2n-1) = n^2$$

vi) Sum of the first ' n ' even +ve integers =

$$2 + 4 + 6 + \dots + 2n = n(n+1)$$

vii) $\sum n^2 \neq (\sum n)^2 \dots \text{etc.},$

viii) $\sum n(n+1) = \frac{n(n+1)(n+2)}{3}$

ix) $\sum n(n+1)(n+2) = \frac{n(n+1)(n+2)(n+3)}{4}$

x) $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots$ 'n' terms
 $= -\frac{n(n+1)}{2}$; if n is even $= \frac{n(n+1)}{2}$; if n is odd

xi) $(n+3)^2 \leq 2^{n+3}$

xii) The sum in the n^{th} bracket of

$1 + (2+3) + (4+5+6) + \dots$ is $\frac{n(n^2+1)}{2}$

xiii) $\sum_{k=1}^n k \left(1 + \frac{1}{n}\right)^{k-1} = n^2$

➤ The inequality

i) $2^n < n!$ is true for all $n \geq 4$

ii) $2^n > 2n+1$ is true for all $n \geq 3$

iii) $2^n \leq (n+1)!$ is true for all $n \in \mathbb{N}$

iv) $2n-3 \leq 2^{n-2}$ is true for all $n \geq 5$

W.E-2: Sum to n terms of $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$ is

Sol: $s_n = \frac{1}{3} \left[1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \dots + \frac{1}{3n-2} - \frac{1}{3n+1} \right]$
 $= \frac{1}{3} \left[1 - \frac{1}{3n+1} \right] = \frac{n}{3n+1}$

➤ $\sin \alpha + \sin(\alpha + \beta) + \sin(\alpha + 2\beta) + \dots + \sin(\alpha + (n-1)\beta)$

$= \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \times \sin\left[\alpha + (n-1)\frac{\beta}{2}\right]$

➤ $\cos \alpha + \cos(\alpha + \beta) + \cos(\alpha + 2\beta) + \dots + \cos(\alpha + (n-1)\beta)$

$= \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \times \cos\left[\alpha + (n-1)\frac{\beta}{2}\right]$

➤ $\cos \alpha \cdot \cos 2\alpha \cdot \cos 4\alpha \dots \cos(2^{n-1}\alpha) = \frac{\sin(2^n \alpha)}{2^n \sin \alpha}$

➤ $\left(1 + \frac{3}{1}\right)\left(1 + \frac{5}{4}\right)\left(1 + \frac{7}{9}\right) \dots \left(1 + \frac{2n+1}{n^2}\right) = (n+1)^2$

➤ For a sequence $T_1, T_2, T_3, T_4, \dots$, the difference of two consecutive terms $T_2 - T_1, T_3 - T_2, T_4 - T_3, \dots$ are in A.P or G.P then n^{th} term of given series are in the form of $an^2 + bn + c$ or $a.r^n + b$ where a,b,c to be determined

SOME IMPORTANT POINTS

i) The sum of cubes of three consecutive natural numbers is always divisible by 9

ii) For all positive integral values of n, $x^n - y^n$ is divisible by $x - y$.

iii) For all positive integral values of n, $x^{2n+1} + y^{2n+1}$ is divisible by $x + y$.

iv) For all positive integral values of n, $x^{2n-1} + y^{2n-1}$ is divisible by $x + y$.

v) $P^{n+1} + (P+1)^{2n-1}$ is divisible by $P^2 + P + 1, \forall n \geq 2$

vi) $n^p - n$ is divisible by $P \forall n \geq 2$ where P is prime.

vii) n is any odd integer then $n(n^2 - 1)$ is divisible by 24.

viii) The product of "n" consecutive natural numbers is always divisible by n!

➤ If x,y,m are positive integers then x is said to be congruent of y modulo m if $x - y$ is divisible by m and is denoted by $x \equiv y \pmod{m}$



C.U.Q

- Mathematical Induction is the principle containing the set**
1) R 2) N 3) Q 4) Z
- Let $P(n)$ be a statement and let $P(n) \Rightarrow P(n+1) \forall n \in N$, then $P(n)$ is true.**
1) for all 'n' 2) for all $n > 1$
3. for all $n > m$, m being a fixed +ve integer
4. Nothing can be said
- The statement $P(n)$**
 $(1 \times 1!) + (2 \times 2!) + (3 \times 3!) + \dots$
 $\dots + (n \times n!) = (n+1)! - 1$ is
1) True for all $n > 1$ 2) Not true for any n 3)
True for all $n \in N$ 4) True for all $n > 10$.
- The smallest +ve integer n for which $n! < \left(\frac{n+1}{2}\right)^n$ holds is**
1) 1 2) 2 3) 3 4) 4
- Let $S(K) = 1 + 3 + 5 + \dots + (2K-1) = K^2 + 3$. Then which of the following is true?**
1) $S(1)$ is correct 2) $S(K) \Rightarrow S(K+1)$
3) $S(K) \neq S(K+1)$
4) P.O.F.M.I can be used to prove the formula
- For every natural number n, $n(n+1)$ is always**
1) even 2) odd
3) multiple of 3 4) multiple of 4
- Let $P(n)$ denote the statement that $n^2 + n$ is odd. It is seen that $P(n) \Rightarrow P(n+1)$, $P(n)$ is true for all**
1) $n > 1$ 2) n 3) $n > 2$ 4) no value of n
- $\{n(n+1)(2n+1) : n \in Z\} \subset$**
1) $\{6k : k \in Z\}$ 2) $\{12k : k \in Z\}$
3) $\{18k : k \in Z\}$ 4) $\{24k : k \in Z\}$
- If $a_n = \sqrt{7 + \sqrt{7 + \sqrt{7 + \dots}}}$ having 'n' radical signs then by methods of mathematical induction which is true**

- 1) $a_n > 7, \forall n \geq 1$ 2) $a_n < 2, \forall n \geq 1$
3) $a_n < 7, \forall n \geq 1$ 4) $a_n > 13, \forall n \geq 1$
- If $a > 0, x > 0$, then**
$$\frac{1}{\sqrt{a} + \sqrt{a+x}} + \frac{1}{\sqrt{a+x} + \sqrt{a+2x}} + \dots$$

$$+ \frac{1}{\sqrt{a+(n-1)x} + \sqrt{a+nx}} =$$

1) $\frac{\sqrt{a+nx} + \sqrt{a}}{x}$ 2) $\frac{\sqrt{a+nx} - \sqrt{a}}{x}$
3) $\frac{x}{\sqrt{a+nx} + \sqrt{a}}$ 4) $\frac{x}{\sqrt{a+nx} - \sqrt{a}}$
- $\forall n \in N, \frac{n^4}{24} + \frac{n^3}{4} + \frac{11n^2}{24} + \frac{n}{4}$ is a**
1) Rational Number 2) Integer
3) Natural Number 4) Real Number
- Let $P(n) : a^n + b^n$ such that a, b are even, then $P(n)$ will be divisible by $a+b$ if**
1) $n > 1$ 2) $\forall n$ is odd
3) $\forall n$ is even 4) $n > 2$
- If p is a prime number, then $n^p - n$ is divisible by p when n is a**
1) Natural number greater than 1
2) Irrational number
3) Complex number 4) Odd number
- The greatest +ve integer which divides $(n+1)(n+2)\dots(n+r)$, for all $n \in N$ is**
1) $(r+1)!$ 2) $r!$ 3) r 4) r-1

C.U.Q-KEY

- 01) 2 02) 4 03) 3 04) 2 05) 2 06) 1
07) 4 08) 1 09) 3 10) 2 11) 3 12) 2
13) 1 14) 2

C.U.Q-HINTS

- By the definition of P.O.F.M.I
- By the definition of P.O.F.M.I
- When $n=1, L.H.S. = 1.1!, R.H.S. = 2! - 1!$
when $n=2, L.H.S. = 1.1! + 2.2!, R.H.S. = 3! - 1!$
Hence $P(n)$ is true $\forall n \in N$

4. By Verification
5. By verification
6. The product of 2 consecutive numbers is always even.
7. It is obvious.
8. Put $n = 1, 2$ and verify the options.
9. $a_1 = \sqrt{7} < 7$; Let $a_m < 7$. Then $a_{m+1} < \sqrt{7 + a_m}$
 $a_{m+1}^2 = 7 + a_m < 7 + 7 = 14$
 $a_{m+1} < \sqrt{14} < 7$; So $a_n < 7, \forall n$
10. By verification
11. Put $n = 1, n = 2$ and verify the options.
12. $P(n) = a^n + b^n \forall n \in N$
 put $n = 1, \therefore P(1) = a + b$ which is divisible by $a + b$
 put $n = 2, \therefore P(2) = a^2 + b^2$ not divisible by $a + b$, put $n = 3$
 $\therefore P(3) = a^3 + b^3 = (a + b)(a^2 - ab + b^2)$
 which is divisible by $a + b$.
 With the help of induction we conclude that $P(n)$ will be divisible by $a + b$ if n is odd.
13. put $n = 4$ and $p = 2$.
14. Product of r successive integers is divisible by $r!$



LEVEL - I (C.W)

PRINCIPLE OF MATHEMATICAL INDUCTION

1. A student was asked to prove a statement by induction. He proved (i) $P(5)$ is true and (ii) truth of $P(n) \Rightarrow$ truth of $P(n+1)$, $n \in N$. On the basis of this, he could conclude that $P(n)$ is true
 - 1) for no $n \in N$ 2) for all $n \geq 5, n \in N$
 - 3) for all $n \in N$ 4) for all $n \geq 1, n \in N$
2. If $P(n)$ be the statement $n(n+1)+1$ is an integer, then which of the following is even
 - 1) $P(2)$ 2) $P(3)$
 - 3) $P(4)$ 4) None of the above
3. $n > 1, n$ even $\Rightarrow$ digit in the units place of $2^{2n} + 1$
 - 1) 5 2) 7 3) 6 4) 1
4. $\log(x)^n = n \cdot \log x$ is true for n .
 - 1) $\forall n \in N$ 2) $\forall n \in Z$
 - 3) n is positive odd integer
 - 4) n is positive even integer

INEQUALITIES

5. $(n!)^2 > n^n$ is true for
 - 1) $\forall n \in N$ 2) $\forall n > 1, n \in N$
 - 3) $\forall n > 2, n \in N$ 4) $\forall n \in Z$
6. Let $P(n): 1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n}$ is true for
 - 1) $\forall n \in N$ 2) $n = 1$
 - 3) $n > 1, \forall n \in N$ 4) $n > 2$

SUMMATION OF SERIES

7. $1^3 - 2^3 + 3^3 - 4^3 + \dots + 9^3 =$
 - 1) 425 2) -425 3) 475 4) -475
8. $\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{8.11} + \dots$ n terms =
 - 1) $\frac{n}{6n+4}$ 2) $\frac{n}{3n+2}$
 - 3) $\frac{n}{4n+6}$ 4) $\frac{1}{2(2n+3)}$
9. $2+7+14+\dots+(n^2+2n-1) =$
 - 1) $\frac{n(2n^2+9n+1)}{6}$ 2) $\frac{2n^2+9n+1}{6}$
 - 3) $\frac{2n^2+9n+1}{12}$ 4) $\frac{2n^2+9n+1}{24}$
10. $1+3+6+10+\dots+\frac{(n-1)n}{2} + \frac{n(n+1)}{2} =$
 - 1) $\frac{n(n+1)(n+2)}{3}$ 2) $\frac{(n+1)(n+2)}{6}$
 - 3) $\frac{n(n+1)(n+2)}{6}$ 4) $\frac{(n+2)(n+1)^2}{3}$
11. $3.6+6.9+9.12+\dots+3n(3n+3) =$
 - 1) $\frac{n(n+1)(n+2)}{3}$ 2) $3n(n+1)(n+2)$
 - 3) $\frac{(n+1)(n+2)(n+3)}{3}$ 4) $\frac{(n+1)(n+2)(n+4)}{4}$

12. $1.6+2.9+3.12+\dots+n(3n+3)=$
 1) $n(n+1)(n+2)$ 2) $(n+1)(n+2)(n+3)$
 3) $(n+2)(n+3)(n+4)$ 4) $(n-1)n(n+1)$
13. $1^3+1^2+1+2^3+2^2+2+3^3+3^2+3+\dots+3n$ terms =
 1) $\frac{n(n+1)(n^2+12n+5)}{12}$
 2) $\frac{n(n+1)(3n^2+7n+8)}{12}$
 3) $\frac{n(n+1)(n+2)(n^2+5n+6)}{12}$
 4) $\frac{(n+1)(n+2)(n+3)}{4}$
14. $\frac{1}{1.3}+\frac{1}{3.5}+\frac{1}{5.7}+\dots+(n-3)$ terms
 1) $\frac{n}{n+2}$ 2) $\frac{n+1}{n(n+5)}$ 3) $\frac{n-3}{2n-5}$ 4) $\frac{n-1}{n(2n-3)}$
15. $1+3+7+15+\dots+n$ terms =
 1) $2^{n+1}-n-2$ 2) n^2+n-2
 3) 2^n+n^2-2 4) n^2+n
16. $2.4+4.7+6.10+\dots+(n-1)$ terms =
 1) $2n^3-2n^2$ 2) $\frac{n^3+3n^2+1}{6}$
 3) $2n^3+2n$ 4) $2n^3-n^2$
17. $1^2+3^2+5^2+\dots$ upto n terms =
 1) $\frac{n(2n-1)(2n+1)}{6}$ 2) $\frac{n(2n-1)(2n+1)}{3}$
 3) $\frac{n(n+1)(2n+1)}{12}$ 4) $\frac{n(n+1)(2n-1)}{6}$
18. Sum to n terms of the series
 $1+(1+x)+(1+x+x^2)+\dots$
 $+ (1+x+x^2+x^3)+\dots$
 1) $\frac{n}{1-x}-\frac{x(1-x^n)}{(1-x)^2}$ 2) $\frac{n}{1-x}+\frac{x(1-x^n)}{(1-x)^2}$
 3) $\frac{n}{1-x}+\frac{x(1+x^n)}{(1-x)^2}$ 4) $\frac{-n}{1-x}+\frac{x(1-x^n)}{(1-x)^2}$
19. If a, b and n are natural numbers then
 $a^{2n-1}+b^{2n-1}$ is divisible by
 1) $a-b$ 2) a^3+b^3 3) $a+b$ 4) a^2+b^2

20. The number $a_n = 6^n - 5n$ for
 $n = 1, 2, 3, \dots$ when divided by 25
 leave the remainder
 1) 9 2) 7 3) 3 4) 1
- DIVISIBILITY**
21. If 'a' and 'b' are natural numbers such that
 $a^2 - b^2$ is prime number then $a^2 - b^2$ equals
 1) $a+b$ 2) $a-b$ 3) ab 4) 1
22. $\forall n \in \mathbb{N}, n^2(n^4-1)$ is divisible by
 1) 60 2) 120 3) 45 4) 90
23. $\forall n \in \mathbb{N}, 49^n + 16n - 1$ is divisible by
 1) 64 2) 49 3) 132 4) 32
24. $\forall n \in \mathbb{N}, 7^{2n} + 3^{n-1} \cdot 2^{3n-3}$ is divisible by
 1) 50 2) 25 3) 2425 4) 2550
25. For all integers $n \geq 1$, which of the following
 is divisible by 9?
 1) $8^n + 1$ 2) $4^n - 3n - 1$
 3) $3^{2n} + 3n + 1$ 4) $10^n + 1$

LEVEL-I (C.W)-KEY

- 01) 2 02) 4 03) 2 04) 3 05) 3 06) 3
 07) 1 08) 1 09) 1 10) 3 11) 2 12) 1
 13) 2 14) 3 15) 1 16) 1 17) 2 18) 1
 19) 3 20) 4 21) 1 22) 1 23) 1 24) 2
 25) 2

LEVEL-I (C.W)-HINTS

- By Verification
- $P(n) = n(n+1)+1, P(2) = 6+1 = 7$
 $P(3) = 3 \times 4 + 1 = 13, P(4) = 4 \times 5 + 1 = 21$
 None of the above is even
- Put $n = 2, 2^4 + 1 = 17$.
- By Verification
- By Verification
- For $n = 1$, L.H.S. of $P(1) = 1$
 R.H.S. of $P(1) = 2 - 1 = 1$, L.H.S. = R.H.S.
 $\therefore 1$ can not be $< 1 \therefore P(1)$ is not true.
 as L.H.S. of $P(1) = 1 < 1$ (false) again $n = 2$
 L.H.S. of $P(2) = 1 + 1/4 = 5/4$
 R.H.S. of $P(2) = 2 - 1/2 = 3/2 = 6/4$
 $\therefore$ L.H.S. $P(2) < 6/4$ of $P(2)$
 $\therefore P(2)$ is true

7. $\sum_{n=1}^9 n^3 - 2(2^3 + 4^3 + 6^3 + 8^3)$
8. Put $n = 2$ and verify the options.
9. Put $n = 2$ and verify the options.
10. Put $n = 2$ and verify the options.
11. Put $n = 2$ and verify the options.
12. Put $n = 2$ and verify the options.
13. Put $n = 2$ and verify the options.
14. Put $n - 3 = 1$, $S = \frac{1}{1.3} = \frac{1}{3}$
15. Put $n = 2$ and verify the options.
16. Put $n = 2$ and verify the options.
17. Put $n = 2$ and verify the options
18. $\frac{1-x}{1-x} + \frac{1-x^2}{1-x} + \frac{1-x^3}{1-x} + \dots + \frac{1-x^n}{1-x}$
19. Put $n = 2$ and verify the options
20. Put $n = 2$ and verify the options
21. We know that $a^2 - b^2 = (a+b)(a-b)$ given
 $a^2 - b^2$ is a prime $\therefore a^2 - b^2$ is either divisible by 1 or the number itself
 $\therefore a^2 - b^2 = a + b$ ($\because a + b \neq 1$ & $a - b = 1$
as $a, b \in N$)
22. Put $n = 2$ and verify the options.
23. Put $n = 1$ and verify the options.
24. Put $n = 1, 2$ and verify the options.
25. Using substitution verify the options



LEVEL - I (H.W)

SUMMATION OF SERIES

1. $1+4+10+19+\dots+\frac{3n^2-3n+2}{2} =$
1) $\frac{n^2(n^2+1)}{2}$ 2) $\frac{n(n^2+1)}{2}$
3) $\frac{n^2(n+1)}{2}$ 4) $\left\{\frac{n(n+1)}{2}\right\}^2$
2. $1.3+3.5+5.7+\dots+(2n-1)(2n+1) =$
1) $\frac{n(4n^2+6n-1)}{3}$ 2) $\frac{n(3n^2+5n+1)}{3}$
3) $\frac{n(5n^2+7n-1)}{3}$ 4) $\frac{n(7n^2-5n+1)}{3}$

3. $1.4+2.5+\dots+n(n+3) =$
1) $\frac{n(n+3)(n+5)}{9}$ 2) $\frac{n(n+1)(n+5)}{3}$
3) $\frac{n(n+5)(n+7)}{6}$ 4) $\frac{n(n+3)(n+9)}{12}$
4. $2+3+5+6+8+9+\dots+2n$ terms =
1) $3n^2+2n$ 2) $4n^2+2n$ 3) $4n^2$ 4) $5n^2+2n$
5. n^{th} term of the series $4+14+30+52+\dots =$
1) $5n - 1$ 2) $2n^2 + 2n$
3) $3n^2+n$ 4) $2n^2+2$
6. $2.3+3.4+4.5+\dots+n$ terms =
1) $\frac{n(n^2+6n+11)}{6}$ 2) $\frac{n(n^2+6n+14)}{9}$
3) $\frac{n(n^2+6n+11)}{3}$ 4) $\frac{n(n^2+6n+17)}{12}$
7. The n^{th} term of the series $3+7+13+21+\dots$ is
1) $4n - 1$ 2) $n^2 + 2n$ 3) $n^2 + n + 1$ 4) $n^2 + 2$
8. The value of
 $1.2.3 + 2.3.4 + 3.4.5 + \dots + n$ terms is
1) $\frac{n(n+1)(n+2)(n+3)}{12}$
2) $\frac{n(n+1)(n+2)(n+3)}{3}$
3) $\frac{n(n+1)(n+2)(n+3)}{4}$
4) $\frac{(n+2)(n+3)(n+4)}{6}$
9. $3.6+4.7+5.8+\dots+(n-2)$ terms =
1) $n^3 + n^2 + n + 2$
2) $\frac{2n^3 + 12n^2 + 10n - 84}{6}$
3) $2n^3 + 12n^2 + 10n + 84$
4) $2n^3 - 12n^2 + 10n - 84$
10. $2 + 3.2 + 4.2^2 + \dots$ upto n terms =
1) $(n+1)2^{n-1}$ 2) $n 2^{n+1}$
3) $n 2^n$ 4) $(n+1)2^n$

11. If $1 + 5 + 12 + 22 + 35 + \dots +$ to n terms $= \frac{n^2(n+1)}{2}$ then n^{th} term of

L.H.S

- 1) $\frac{n(4n-1)}{3}$ 2) $\frac{n(3n-1)}{2}$
3) $\frac{n(3n+1)}{2}$ 4) $\frac{n(4n+1)}{3}$

12. $2.4 + 4.7 + 6.10 + \dots$ upto $(n-1)$ terms =

- 1) $2n^3 + 2n^2$ 2) $\frac{1}{6}(n^3 + 3n^2 + 1)$
3) $2n^3 + 2n$ 4) $2n^3 - 2n^2$

13. $\frac{1}{4.7} + \frac{1}{7.10} + \frac{1}{10.13} + \dots$ upto n terms =

- 1) $\frac{n}{4(4n+3)}$ 2) $\frac{n}{4(6n+1)}$
3) $\frac{n}{4(3n+4)}$ 4) $\frac{n}{4(3n-4)}$

DIVISIBILITY

14. Sum of the cubes of three successive natural number is divisible by

- 1) 9 2) 27 3) 54 4) 99

15. $\forall n \in \mathbb{N}, 3^{2n} + 7$ is divisible by

- 1) 8 2) 16 3) 24 4) 64

16. $\forall n \in \mathbb{N}, 7.5^{2n} + 12.6^n$ is divisible by

- 1) 13 2) 19 3) 247 4) 26

17. For every natural number n , $3^{2n+2} - 8n - 9$ is divisible by

- 1) 16 2) 128 3) 256 4) 512

18. For every integer $n \geq 1, (3^{2^n} - 1)$ is divisible by

- 1) 2^{n+2} 2) 2^{n+1} 3) 2^{n+3} 4) 2^n

19. Use mathematical induction, to find

$41^n - 14^n$ is a multiple of

- 1) 2, 8 2) 27 3) 26 4) 14

LEVEL - I (H.W)-KEY

- 01) 2 02) 1 03) 2 04) 1 05) 3 06) 3
07) 3 08) 3 09) 2 10) 3 11) 2 12) 4
13) 3 14) 1 15) 1 16) 2 17) 1 18) 1 19) 2

LEVEL - I (H.W)-HINTS

1. Put $n = 2$ and verify the options.
2. Put $n = 2$ and verify the options.
3. Put $n = 2$ and verify the options.
4. Put $n = 1$ and verify the options.

$$\text{L.H.S} = 2 + 3$$

$$\text{R.H.S} = 3(1)^2 + 2(1) = 5$$

5. Given $T_2 = 14$
Put $n = 2$ and verify the options.
6. Put $n = 2$ and verify the options.
7. Put $n = 3$ and verify the options.
8. By verification
9. Put $n = 3$ and verify the options.
10. Put $n = 2$ and verify the options
11. Put $n = 2$ verify the options
12. Put $n - 1 = 2$ and verify the options

$$13. \frac{1}{3} \left[\frac{1}{4} - \frac{1}{7} + \frac{1}{7} - \frac{1}{10} + \dots + \frac{1}{3n+1} - \frac{1}{3n+4} \right]$$

$$= \frac{n}{4(3n+4)}$$

14. $1^3 + 2^3 + 3^3 = 36$ divisible by 9.
15. Put $n = 1, 3^2 + 7 = 16$
Put $n = 2, 3^4 + 7 = 88$, G.C.D. of 16 & 88 = 8
16. Put $n = 1, 7.5^2 + 12.6 = 247$
Put $n = 2, 7.5^4 + 12.6^2 = 4807$
G.C.D. of 247 & 4807 = 19
17. $3^{2n+2} - 8n - 9, \forall n \in \mathbb{N}$
put $n = 2, f(2) = 704$, divisible by 16
18. Let $P(n) = 3^{2^n} - 1$
 $P(1) = 3^{2^1} - 1 = 8 = 1.8$ is divisible by 2^3 .
 $P(2) = 3^{2^2} - 1 = 80 = 5.2^4$ is divisible by 2^4 .
19. Verification

 LEVEL - II (C.W)
**FINITE MATHEMATICAL INDUCTION,
SUMMATION OF SERIES**

- If $2^3 + 4^3 + 6^3 + \dots + (2n)^3 = Kn^2(n+1)^2$ then $k =$
1) $1/2$ 2) 1 3) $3/2$ 4) 2
- If $a_k = \frac{1}{k(k+1)}$ for $k = 1, 2, 3, \dots, n$, then
 $\left(\sum_{k=1}^n a_k\right)^2 =$
1) $\frac{n}{n+1}$ 2) $\frac{n^2}{(n+1)^2}$ 3) $\frac{n^4}{(n+1)^4}$ 4) $\frac{n^6}{(n+1)^6}$
- $\sum_{n=0}^{\infty} (-1)^n x^{n+1} =$
1) $\frac{x^n}{2(1+x)}$ 2) $\frac{x}{1+x}$ 3) $\frac{x}{x-1}$ 4) $\frac{x^n}{x-1}$
- $\cos \theta + \cos 2\theta + \cos 3\theta + \dots + \cos \{(n-1)\theta\} + \cos n\theta =$
1) $\frac{\cos \left\{ \frac{1}{2}(n+1)\theta \right\} \cdot \sin \left(\frac{n\theta}{2} \right)}{\sin \left(\frac{\theta}{2} \right)}$ 2) $\frac{\cos(n+1)\theta}{\sin \left(\frac{\theta}{2} \right)}$
3) $\frac{\cos \left(\frac{(n-1)\theta}{2} \right) \sin \left(\frac{n\theta}{2} \right)}{\sin \frac{\theta}{2}}$
4) $\frac{\sin \left(\frac{n\theta}{2} \right) \cdot \cos \{(n+1)\theta\}}{\sin \left(\frac{\theta}{2} \right)}$
- If 'n' is a positive integer, then
 $n.1 + (n-1).2 + (n-2).3 + \dots + 1.n =$
1) $\frac{n(n+1)}{2}$ 2) $\frac{n(n+1)(n+2)}{6}$
3) $\frac{(n+1)(n+2)}{2}$ 4) $\frac{n(n+1)(2n+1)}{6}$

MATHEMATICAL INDUCTION

- $1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots + n$
brackets =
1) $\frac{n(n+1)^2(n+2)^2}{12}$ 2) $\frac{n(n+1)^2(n+2)}{12}$
3) $\frac{n^2(n+1)(n+2)}{12}$ 4) $\frac{(n+1)}{2}$
1) $\frac{1}{2} \cdot \frac{2}{2} + \frac{2}{2} \cdot \frac{3}{2} + \frac{3}{2} \cdot \frac{4}{2} + \dots$ n terms
- $\frac{2}{1^3} \cdot \frac{2}{2} + \frac{2}{1^3+2^3} \cdot \frac{3}{2} + \frac{3}{1^3+2^3+3^3} \cdot \frac{4}{2} + \dots$ n terms
1) $\frac{n^2}{(n+1)^2}$ 2) $\frac{n^3}{(n+1)^3}$ 3) $\frac{n}{n+1}$ 4) $\frac{1}{n+1}$
- If $t_n = \sum_1^n n$, then $t_n^1 = \sum_1^n t_n =$
1) $\frac{n(n+1)}{2}$ 2) $\frac{n(n+3)}{2}$
3) $\frac{n(n+1)(n+2)}{6}$ 4) $\frac{n(n+4)}{3}$
- Let the statement $m^2 > 100$, the statement $P(k+1)$ will be true if
1) $P(1)$ is true 2) $P(2)$ is true
3) $P(k)$ is true 4) none of these
- For any integer $n \geq 1$, then
 $\sum_{k=1}^n k(k+2)$ is equal to [Eam-2008]
1) $\frac{n(n+1)(n+2)}{6}$ 2) $\frac{n(n+1)(2n+1)}{6}$
3) $\frac{n(n+1)(2n+7)}{6}$ 4) $\frac{n(n+1)(2n+9)}{6}$
- $1 + \frac{1}{2}(1+2) + \frac{1}{3}(1+2+3) + \frac{1}{4}(1+2+3) + \dots$
upto 20 terms is
1) 110 2) 111 3) 115 4) 116
- If $\frac{1}{2 \times 4} + \frac{1}{4 \times 6} + \frac{1}{6 \times 8} + \dots + n$ terms $= \frac{kn}{n+1}$
then $k =$ [Eam-2012]
1) $\frac{1}{4}$ 2) $\frac{1}{2}$ 3) 1 4) $\frac{1}{8}$

DIVISIBILITY

13. The greatest positive integer which divides $n(n+1)(n+2)\dots(n+r-1)$, $\forall n \in N$ is
 1) $r!$ 2) $(r+1)!$ 3) $n+r$ 4) $n-r+1$
14. If $10^n + 3.4^n + x$ is divisible by 9 for all $n \in N$, then least positive value of 'x' is
 1) 1 2) 5 3) 14 4) 23

INEQUALITIES

15. If $n > 1$ then
 1) $\frac{(2n)!}{(n!)^2} = \frac{4n}{2n+1}$ 2) $\frac{(2n)!}{(n!)^2} < \frac{4n}{2n+1}$
 3) $\frac{(2n)!}{(n!)^2} > \frac{4n}{2n+1}$ 4) none
16. $\forall n \in N$,
 1) $|\sin(nx)| < |\sin x|$ 2) $|\sin(nx)| < n|\sin x|$
 3) $|\sin(nx)| \leq n|\sin x|$ 4) $\sin(nx) \leq \sin n$

LEVEL - II (C.W)-KEY

- 01) 4 02) 2 03) 2 04) 1 05) 2 06) 2
 07) 3 08) 3 09) 3 10) 3 11) 3 12) 1
 13) 1 14) 2 15) 3 16) 3

LEVEL - II (C.W)-HINTS

- $2^3(1^3 + 2^3 + 3^3 + \dots + n^3) = kn^2(n+1)^2$
- Put $n = 1$ and verify the options.
- $x(1-x+x^2-x^3+\dots) = x(1+x)^{-1}$
- Put $n = 1$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- Put $n = 2$ and verify the options.
- $\sum_{i=1}^n t_n = \frac{1}{2}[\sum n^2 + \sum n]$
- $P(r)$ is true
 $\Rightarrow r^2 > 100 \Rightarrow r^2 + 2r + 1 > 100 + 2r + 1$
 $\Rightarrow (r+1)^2 > 100 \Rightarrow P(k+1)$ is true. $P(k+1)$ is true when every $P(k)$ is so.
- Put $n = 2$ and verify the options
- $t_n = \frac{1}{n}(1+2+3+\dots+n) = \frac{1}{n} \frac{n(n+1)}{2} = \frac{n+1}{2}$
 Now find $\sum_{i=1}^n \left(\frac{n+1}{2}\right) = \frac{1}{2} \frac{n(n+1)}{2} + \frac{n}{2} = S_n$
 now put $n = 20$

12. $\frac{kn}{n+1} = \left[\frac{1}{2.4} + \frac{1}{4.6} + \frac{1}{6.8} + \dots n \text{ terms} \right]$
 $= \frac{1}{2} \left[\frac{4-2}{2.4} + \frac{6-4}{4.6} + \frac{8-6}{6.8} + \dots + \frac{1}{2n(2n+2)} \right]$
 $= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{4} + \frac{1}{4} - \frac{1}{6} + \frac{1}{6} - \frac{1}{8} + \dots + \frac{1}{2n} - \frac{1}{2n+2} \right]$
 $= \frac{1}{2} \left[\frac{1}{2} - \frac{1}{2(n+1)} \right] = \frac{n}{4(n+1)} \Rightarrow k = \frac{1}{4}$
13. Let $P(n) = n(n+1)(n+2)\dots(n+r-1)$
 $= (n+0)(n+1)(n+2)\dots(n+(r-1))$
 $= \text{Product of } r \text{ consecutive integers}$
14. $n = 1 \Rightarrow 10 + 3.4 + x = 9m \Rightarrow x = 5$
15. If $n = 2$ then $\frac{(2n)!}{(n!)^2} = \frac{24}{4} = 6, \frac{4n}{2n+1} = \frac{8}{3}$
 $\Rightarrow \frac{(2n)!}{(n!)^2} > \frac{4n}{2n+1}$

16. $|\sin nx| \leq 1 \leq n|\sin x|$

LEVEL - II (H.W)

- $4^3 + 5^3 + 6^3 + \dots + 10^3 =$
 1) 1905 2) 2358 3) 2447 4) 2989
- $\left(\sum n^3\right)\left(\sum n\right) = \left(\sum n^2\right)^2$ if
 1) $n = 3$ 2) $n = 1$ 3) $n^2 = 3$ 4) $n = -1$
- $\sum_{k=1}^n k \left(1 + \frac{1}{n}\right)^{k-1} =$
 1) $n(n-1)$ 2) $n(n+1)$ 3) n^2 4) $(n+1)^2$
- Sum of n^{th} bracket of**
(1) + (2+3+4) + (5+6+7+8+9) + is
 1) $(n-1)^3 + n^3$ 2) $(n-1)^3 + 8n^2$
 3) $\frac{(n+1)(n+2)}{6}$ 4) $\frac{(n+3)(n+2)}{12}$
- $\frac{1^2}{1} + \frac{1^2+2^2}{1+2} + \frac{1^2+2^2+3^2}{1+2+3} + \dots + n \text{ terms} =$
 1) $\frac{n(n+3)}{4}$ 2) $\frac{n(n+3)}{5}$ 3) $\frac{n(n+2)}{3}$ 4) $\frac{n(n+5)}{6}$

6. $\sum_{k=1}^5 \frac{1^3 + 2^3 + \dots + k^3}{1 + 3 + 5 + \dots + (2k-1)} =$
 1) 22.5 2) 24.5 3) 28.5 4) 32.5
7. $S_n = 1^3 + 2^3 + 3^3 + \dots + n^3$ and
 $T_n = 1 + 2 + 3 + 4 + \dots + n$ [Eam-2007]
 1) $S_n = T_n^3$ 2) $S_n = T_n^2$ 3) $S_n = T_n^2$ 4) $S_n = T_n^3$
8. A = sum of first 10 natural numbers,
 B = sum of squares of first 10
 natural numbers, C = sum of cubes of first 10
 natural numbers, D = sum of first 10 even
 natural numbers, then increasing order of
 A, B, C, D
 1) A, B, C, D 2) A, C, D, B
 3) A, B, D, C 4) A, D, B, C
9. If $\sum_{r=1}^n t_r = \frac{1}{12} n(n+1)(n+2)$ then $\sum_{r=1}^n \frac{1}{t_r}$
 1) $\frac{2n}{n+1}$ 2) $\frac{4n}{n+1}$ 3) $\frac{3n}{n+2}$ 4) $\frac{3n}{n+1}$
10. $1 + (1+3) + (1+3+5) + \dots$ n brackets =
 1) $\frac{n(n+1)(n+2)}{6}$
 2) $\frac{n(n+1)(3n^2 + 23n + 46)}{12}$
 3) $\frac{n(27n^3 + 90n^2 + 45n - 50)}{4}$
 4) $\frac{n(n+1)(2n+1)}{6}$
11. $\frac{1}{1^3} + \frac{1+2}{1^3+2^3} + \frac{1+2+3}{1^3+2^3+3^3} + \dots$ upto n
 terms =
 1) $\frac{n}{n+1}$ 2) $\frac{n}{2(n+1)}$ 3) $\frac{2n}{n+1}$ 4) $\frac{2}{n(n+1)}$
- DIVISIBILITY**
12. $\forall n \in N, 5^{2n+2} - 24n - 25$ is divisible by
 1) 576 2) 25 3) 24 4) 50

13. The remainder left out when $8^{2n} - (62)^{2n+1}$ is
 divided by 9 is
 1) 2 2) 7 3) 8 4) 0

INEQUALITIES

14. $\frac{(2n)!}{2^{2n}(n!)^2} \leq$
 1) $\frac{1}{\sqrt{3n+1}}$ 2) $\frac{1}{\sqrt{3n+2}}$ 3) $\frac{1}{\sqrt{3n+4}}$ 4) $\frac{1}{\sqrt{3n+5}}$
15. If $n > 1$ then
 1) $n! = \left(\frac{n+1}{2}\right)^n$ 2) $n! < \left(\frac{n+1}{2}\right)^n$
 3) $n! > \left(\frac{n+1}{2}\right)^n$ 4) none

LEVEL-II (H.W) KEY

- 01) 4 02) 2 03) 3 04) 1 05) 3 06) 1
 07) 3 08) 4 09) 2 10) 4 11) 3 12) 1
 13) 1 14) 1 15) 2

LEVEL-II (H.W) HINTS

1. $\sum_{n=1}^{10} n^3 - \sum_{r=1}^3 r^3$
2. Put $n = 1$ and verify the options.
3. By verification.
4. Put $n = 2$ and verify the options.
5. Put $n = 2$ and verify the options.
6. $\sum_{k=1}^5 \left(\frac{k^2(k+1)^2}{4k^2} \right) = \sum_{k=1}^5 \left(\frac{(k+1)^2}{4} \right)$
7. Put $n = 2$ and verify the options.
8. $A = \sum n, B = \sum n^2, C = \sum n^3, D = \sum 2n$
9. $t_r = \sum_{k=1}^r t_k - \sum_{k=1}^{r-1} t_k = \frac{1}{12} r(r+1)(r+2) -$
 $\frac{1}{12} (r-1)r(r+1) = \frac{r(r+1)}{4}$
10. Conceptual
11. $t_n = \frac{\sum n}{\sum n^2} = \frac{2}{n(n+1)}$

12. Put $n = 1$ and verify the options.
 13. Using modulo arithmetic
 $8 \equiv -1 \pmod{9}$ also $62 \equiv -1 \pmod{9}$
 $\Rightarrow 8^{2n} - (62)^{2n+1} = [(-1)^{2n} - (-1)^{2n+1}] \pmod{9}$
 $= (1+1) \pmod{9} = 2 \pmod{9} \Rightarrow \text{Remainder} = 2$
14. If $n = 1$ then, $\frac{(2n)!}{2^n (n!)^2} = \frac{2}{4} = \frac{1}{2}$,
 $\frac{1}{\sqrt{3n+1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}, \frac{1}{\sqrt{3n+2}} = \frac{1}{\sqrt{5}},$
 $\frac{1}{\sqrt{3n+4}} = \frac{1}{\sqrt{7}}$
 $\frac{1}{\sqrt{3n+5}} = \frac{1}{\sqrt{8}}, \therefore \frac{(2n)!}{2^n (n!)^2} \leq \frac{1}{\sqrt{3n+1}}$
15. If $n = 2$ then $n! = 2, \left(\frac{n+1}{2}\right)^n = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$
 $\Rightarrow n! < \left(\frac{n+1}{2}\right)^n$



LEVEL - III

1. $\frac{1.2^2 + 2.3^2 + 3.4^2 + \dots + n(n+1)^2}{1^2.2 + 2^2.3 + 3^2.4 + \dots + n^2(n+1)} =$
 1) $\frac{3n+1}{3n+5}$ 2) $\frac{3n+5}{3n+1}$
 3) $(3n+1)(3n+5)$ 4) $\frac{3n+5}{3n+7}$
2. If $P(n) = 1 + 2 + 3 + \dots + n$ is a perfect square
 N^2 and N is less than 100, then possible
 values of n are
 1) only 1 2) 1 & 8 3) only 8 4) 1, 8, 49
3. For any $n \in N$, the value of the expression
 $\sqrt{2 + \sqrt{2 + \sqrt{2 + \dots n \text{ times}}}}$ is
 1) $2 \cos\left(\frac{\pi}{2^{n+1}}\right)$ 2) $2 \sin\left(\frac{\pi}{2^{n+1}}\right)$
 3) $\sqrt{2} \cos(2^{n+1}\pi)$ 4) $2 \cos(2^n \pi)$

4. $7 + 77 + 777 + \dots + (777 \dots 7 \text{ n times}) =$
 1) $\frac{7}{81}(10^{n+1} - 9n - 10)$ 2) $\frac{7}{81}(10^n - 9n - 10)$
 3) $\frac{7}{81}(10^{n+1} + 9n + 10)$ 4) $\frac{7}{81}(10^{n+1} + 9n - 10)$
5. $\forall n \in N, 1 + 2x + 3x^2 + \dots + n.x^{n-1} =$
 $(x \in R, x \neq 1)$
 1) $\frac{1 - (n+1)x^n + n.x^{n+1}}{(1-x)^2}$ 2) $\frac{(n+1)x^n}{(1-x)^2}$
 3) $\frac{1 - (n+1)x^n + n.x^{n+1}}{(1+x)^2}$ 4) $\frac{(n-1)x^n}{(1+x)^2}$
6. If the sum to 'n' terms of an A.P. is $\frac{4n^2 - 3n}{4}$,
 then the n^{th} term of the A.P. is
 1) $\frac{5n-1}{4}$ 2) $\frac{8n-7}{4}$ 3) $\frac{3n^2-2}{4}$ 4) $\frac{7n-8}{4}$
7. $1 + \left(\frac{1}{3} + \frac{1}{3^2}\right) + \left(\frac{1}{3^3} + \frac{1}{3^4} + \frac{1}{3^5}\right) + \dots$ sum of the
 terms in the n^{th} bracket =
 1) $\frac{(3^n - 1)^3}{2.4^{n-1}}$ 2) $\frac{3^n - 1}{2.3^{(n-1)(n+2)/2}}$ 3) $\frac{3^n + 1}{3.7^{n-1}}$ 4) $\frac{3^{n-1}}{2^{n+1}}$
8. $(1+x)^n - nx - 1$ is divisible by (where $n \in N$)
 1) $2x$ 2) x^2 3) $2x^3$ 4) all of these
9. For a positive integer ($n > 1$),
 Let $a(n) = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{(2^n) - 1}$. Then
 1) $a(100) < 100$ 2) $a(100) > 100$
 3) $a(20) = 100$ 4) $a(200) \geq 100$
10. The sum of the first 'n' terms of the series
 $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$ is
 $\frac{n(n+1)^2}{2}$ when n is even. When n is odd the
 sum is
 1) $\frac{3n(n+1)}{2}$ 2) $\frac{n^2(n+1)}{2}$
 3) $\frac{n(n+1)^2}{4}$ 4) $\left[\frac{n(n+1)}{2}\right]^2$

11. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}, \forall n \in \mathbb{N}$,
then $H_1 + H_2 + H_3 + \dots + H_n =$
1) $(n+1)H_n - n$ 2) $(n+1)H_n + n$
3) $(n+1)H_n$ 4) $(n-1)H_n - n$
12. $S_n = \frac{1+2+3+\dots+n}{n}$ then
 $S_1^2 + S_2^2 + S_3^2 + \dots + S_n^2 =$
1) $\frac{n}{24}(2n^2 + 9n + 13)$ 2) $\frac{1}{24}(2n^2 + 9n + 13)$
3) $\frac{n^2}{24}(2n^2 + 9n + 13)$ 4) $\frac{n}{24}(2n^2 - 9n + 13)$
13. If $t_n = \frac{1}{4}(n+2)(n+3)$ for $n = 1, 2, 3, \dots$
then $\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \dots + \frac{1}{t_{2003}} =$
1) $\frac{4006}{3006}$ 2) $\frac{4003}{3007}$ 3) $\frac{4006}{3008}$ 4) $\frac{4006}{3009}$
14. The value of the sum in the 50th bracket of
 $(1) + (2+3) + (4+5+6) + (7+8+9+10) + \dots$ is
1) 62525 2) 65225 3) 56255 4) 55625
15. $1 + \frac{x}{a_1} + \frac{x(x+a_1)}{a_1 a_2} + \dots +$
 $\frac{x(x+a_1)(x+a_2)\dots(x+a_{n-1})}{a_1 a_2 \dots a_n} =$
1) $\frac{(x+a_1)(x+a_2)\dots(x+a_n)}{a_1 a_2 \dots a_n}$
2) $\frac{(x-a_1)(x-a_2)\dots(x-a_n)}{a_1 a_2 \dots a_n}$
3) $(x+a_1)(x+a_2)\dots(x+a_n)$
4) $(x-a_1)(x-a_2)\dots(x-a_n)$
16. For all positive integers $n > 1$,
 $\{x(x^{n-1} - n.a^{n-1}) + a^n(n-1)\}$ is divisible by
1) $(x-a)^2$ 2) $x-a$ 3) $2(x-a)$ 4) $x+a$

17. $\forall n \in \mathbb{N}, x \in \mathbb{R}$,
 $\tan^{-1}\left[\frac{x}{1.2+x^2}\right] + \tan^{-1}\left[\frac{x}{2.3+x^2}\right] + \dots +$
 $\tan^{-1}\left[\frac{x}{n(n+1)+x^2}\right] =$
1) $\tan^{-1}\left[\frac{x}{n}\right] - \tan^{-1}\left[\frac{x}{n+1}\right]$
2) $\tan^{-1}[x] - \tan^{-1}\left[\frac{x}{n+1}\right]$
3) $\tan^{-1}[n+1] - \tan^{-1}[x]$ 4) $\tan^{-1}[x]$
18. $\tan^{-1}\left(\frac{1}{1+1+1^2}\right) + \tan^{-1}\left(\frac{1}{1+2+2^2}\right)$
 $+ \dots + \tan^{-1}\left(\frac{1}{1+n+n^2}\right) =$
1) $\tan^{-1}(n+1) + \pi$ 2) $\tan^{-1}(n+1) + \frac{\pi}{4}$
3) $\tan^{-1}(n+1)$ 4) $\tan^{-1}(n+1) - \frac{\pi}{4}$
19. $\frac{1}{2}\tan\left(\frac{x}{2}\right) + \frac{1}{4}\left(\tan\frac{x}{4}\right) + \dots + \frac{1}{2^n}\tan\left(\frac{x}{2^n}\right) =$
1) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right)$ 2) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) + \cot x$
3) $\frac{1}{2^n}\cot\left(\frac{x}{2^n}\right) - \cot x$ 4) $\cot\left(\frac{x}{2^n}\right) - \cot x$
20. Sum of first 'n' terms of the series
 $\frac{3}{2} + \frac{5}{4} + \frac{9}{8} + \frac{17}{16} + \dots$
1) $n-1+2^{-n}$ 2) $n+1+2^{-n}$
3) $n+1-2^{-n}$ 4) $n-1-2^{-n}$
21. If n is even, then the sum of first 'n' terms of
the series $1^2 + 2.2^2 + 3^2 + 2.4^2 + 5^2 + 2.6^2 + \dots$
is
1) $\frac{n(n+1)^2}{3}$ 2) $\frac{n(n+1)^2}{4}$
3) $\frac{n(n+1)^2}{2}$ 4) $\frac{n^2(n+1)}{2}$

22. If $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$, then the value of

$$1 + \frac{3}{2} + \frac{5}{3} + \dots + \frac{2n-1}{n} \text{ is}$$

- 1) $H_n + n$ 2) $2n - H_n$
3) $n - 1 + H_n$ 4) $H_n + 2n$

23. If $S_1 = \{2\}$, $S_2 = \{3, 6\}$, $S_3 = \{4, 8, 16\}$,
 $S_4 = \{5, 10, 20, 40\}$, then the sum of

numbers in the set S_{15} is

- 1) $5(2^{15})$ 2) $16(2^{15} - 1)$
3) $16(2^{16} - 1)$ 4) $15(2^{15} - 1)$

24. The sum of first 'n' terms of the series
 $1^2 + (1)(2) + 3^2 + (3)(4) + 5^2 + (5)(6) + 7^2 + \dots$
when n is odd is

- 1) $\frac{1}{12}(n+1)(4n^2 - n + 3)$
2) $\frac{1}{12}n(4n^2 + 3n - 4)$
3) $\frac{1}{6}(n+1)(4n^2 - n + 5)$
4) $\frac{1}{6}n(4n^2 + 5n - 6)$

25. The sets $S_1, S_2, S_3, \dots$ are given by

$$S_1 = \left\{ \frac{2}{1} \right\}, S_2 = \left\{ \frac{3}{2}, \frac{5}{2} \right\},$$

$$S_3 = \left\{ \frac{4}{3}, \frac{7}{3}, \frac{10}{3} \right\}, S_4 = \left\{ \frac{5}{4}, \frac{9}{4}, \frac{13}{4}, \frac{17}{4} \right\}, \dots$$

then the sum of the numbers in the set S_{25} is

- 1) 322 2) 324 3) 325 4) 326

26. If $S_1 = \{2\}$, $S_2 = \left\{ \frac{3}{2}, \frac{4}{2} \right\}$, $S_3 = \left\{ \frac{4}{4}, \frac{5}{4}, \frac{6}{4} \right\}$,

$$S_4 = \left\{ \frac{5}{8}, \frac{6}{8}, \frac{7}{8}, \frac{8}{8} \right\}, \dots \text{ then the sum of}$$

numbers in S_{20} is

- 1) $\frac{1220}{2^{20}}$ 2) $\frac{1563}{2^{19} - 1}$ 3) $\frac{1445}{2^{18}}$ 4) $\frac{1576}{2^{21}}$

27. Sum of the series

$$S = 1^2 - 2^2 + 3^2 - 4^2 + \dots - 2002^2 + 2003^2 \text{ is}$$

- 1) 2007006 2) 1005004
3) 2000506 4) none

28. The positive integer 'n' for which

$$2 \times 2^2 + 3 \times 2^3 + 4 \times 2^4 + \dots + n \times 2^n = 2^{n+10} \text{ is}$$

- 1) 510 2) 511 3) 512 4) 513

29. $\sum_{k=1}^{2n+1} (-1)^{k-1} k^2 =$

- 1) $(n+1)(2n+1)$ 2) $(n+1)(2n-1)$
3) $(n-1)(2n-1)$ 4) $(n-1)(2n+1)$

DIVISIBILITY

30. Let $P(n) = 5^n - 2^n$, $P(n)$ is divisible by 3λ
where λ and n both are odd positive integers
then the least value of 'n' and λ will be

- 1) 13 2) 11 3) 1 4) 5

31. The least remainder when 17^{30} is divided by
5 is

- 1) 1 2) 2 3) 3 4) 4

32. When 2^{301} is divided by 5, the least +ve
remainder is

- 1) 4 2) 8 3) 2 4) 6

33. The remainder when 5^{99} is divided by 13 is

- 1) 6 2) 8 3) 9 4) 10

LEVEL - III-KEY

- 01) 2 02) 4 03) 1 04) 1 05) 1 06) 2
07) 2 08) 2 09) 1 10) 2 11) 1 12) 1
13) 4 14) 1 15) 1 16) 1 17) 2 18) 4
19) 3 20) 3 21) 3 22) 2 23) 2 24) 1
25) 4 26) 1 27) 1 28) 4 29) 1 30) 3
31) 4 32) 3 33) 2

LEVEL - III-HINTS

- Put $n = 2$ and verify the options.
- By Observation
- By Verification
- Put $n = 1$ and verify the options.
- Put $n = 2$ and verify the options.
- $t_n = S_n - S_{n-1}$
- By Verification
- Put $n = 2$, and $x = 3$
not divisible by 6, 54 but divisible by 9.
- $\frac{n}{2} < a(n) < n$

10. By verification
11. By Verification
12. Put $n = 2$ and verify the options.

$$S_1^2 + S_2^2 = 1 + \frac{9}{4} = \frac{13}{4}$$

$$13. \frac{1}{t_n} = \frac{4}{(n+2)(n+3)} = 4 \left[\frac{1}{(n+2)} - \frac{1}{(n+3)} \right]$$

$$14. \text{First term of 50th bracket} \\ = (1+2+3+\dots+49)+1 = 1226$$

15. Put $n = 1$ and verify the options.
16. Put $n = 2$ and verify the options.
17. Put $n = 1$ and verify the options.

$$18. \tan^{-1} \left(\frac{1}{1+n+n^2} \right) = \tan^{-1}(n+1) - \tan^{-1}(n)$$

$$\tan^{-1} \left(\frac{1}{1+1+1^2} \right) + \tan^{-1} \left(\frac{1}{1+2+2^2} \right)$$

$$+ \dots + \tan^{-1} \left(\frac{1}{1+n+n^2} \right)$$

$$= \tan^{-1}(n+1) - \tan^{-1}(1) = \tan^{-1}(n+1) - \frac{\pi}{4}$$

19. Put $n = 1$ and verify the options.
20. Put $n = 2$ and verify the options.
21. By verification

$$22. (2-1) + \left(2 - \frac{1}{2}\right) + \left(2 - \frac{1}{3}\right) + \dots$$

$$= 2n - \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots\right) = 2n - H_n$$

$$23. T_n = (n+1)(2^n - 1)$$

$$24. \text{Put } n = 3 \text{ and verify}$$

$$25. S_{25} = \left\{ \frac{26}{25}, \frac{51}{25}, \dots, 25 \text{ terms} \right\}$$

$$26. \text{Ist term in } S_1 = \frac{2}{2^0}, \quad \text{Ist term in } S_2 = \frac{3}{2^1},$$

$$\text{Ist term in } S_3 = \frac{4}{2^2}, \quad \text{Ist term in } S_{20} = \frac{21}{2^{19}}$$

$$\text{Sum in } S_{20} = \frac{20}{2} \left[2 \times \frac{21}{2^{19}} + (20-1) \frac{1}{2^{19}} \right]$$

$$27. \text{We can write S as}$$

$$S = (1-2)(1+2) + (3-4)(3+4) + \dots +$$

$$(2001-2002)(2001+2002) + 2003^2$$

$$= -[1+2+3+4+\dots+2002] +$$

$$2003^2 = -\frac{1}{2}(2002)(2003) + 2003^2 = 2007006$$

$$28. \text{We have } 2^{n+10} = 2 \cdot 2^2 + 3 \cdot 2^3 + 4 \cdot 2^4 + \dots +$$

$$n \cdot 2^n \Rightarrow 2(2^{n+10}) = 2 \cdot 2^3 + 3 \cdot 2^4 + \dots +$$

$$+(n-1) \cdot 2^n + n \cdot 2^{n+1}$$

subtracting, we get

$$-2^{n+10} = 2 \cdot 2^2 + 2^3 + 2^4 + \dots + 2^n - n \cdot 2^{n+1} = 8$$

$$+ \frac{8(2^{n-2}-1)}{2-1} - n \cdot 2^{n+1}$$

$$= 8 + 2^{n+1} - 8 - n \cdot 2^{n+1} = 2^{n+1} - (n)2^{n+1}$$

$$\Rightarrow 2^{10} = 2n - 2 \Rightarrow n = 513$$

$$29. 1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + (2n+1)^2$$

$$= 1^2 + 2^2 + 3^2 + \dots + (2n+1)^2$$

$$- 2 \left[2^2 + 4^2 + \dots + (2n)^2 \right]$$

$$= \frac{(2n+1)(2n+2)(4n+3)}{6}$$

$$- \frac{8 \cdot n(n+1)(2n+1)}{6} = (n+1)(2n+1)$$

$$30. P(n) = 5^n - 2^n$$

$$n = 1, \therefore P(1) = 3\lambda = 3 \therefore \lambda = 1$$

$$\text{Similarly } n = 5, \therefore P(5) = 5^5 - 2^5 \\ = 3125 - 32 = 3093 = 3 \times 1031$$

In this case $\lambda = 1031$

Similarly we can check the result for other cases and find that the least value of λ and n is 1.

$$31. 17 \equiv 2 \pmod{5}$$

$$17^5 \equiv 2^5 \pmod{5} \equiv 2 \pmod{5}$$

$$\Rightarrow (17^5)^6 = 2^6 \pmod{5} \Rightarrow (17)^{30} \equiv 4 \pmod{5}$$

$$32. 2^4 \equiv 1 \pmod{5} \Rightarrow (2^4)^{75} \equiv (1)^{75} \pmod{5} \text{ i.e.,}$$

$$2^{300} \equiv 1 \pmod{5} \Rightarrow 2^{300} \times 2 \equiv (1 \cdot 2) \pmod{5}$$

$$\Rightarrow 2^{301} \equiv 2 \pmod{5} \therefore \text{Least +ve remainder is 2.}$$

$$33. 5^{99} = (5)(5^2)^{49} = 5(25)^{49} = 5(26-1)^{49}$$

$$= 5 \times 26 \times (\text{positive term}) - 5,$$

so when it is divided by 13 its gives the remainder

'-5' or $(13-5)$ i.e., 8.

LEVEL - IV

- Statement - 1:** For all $n \in N$, $x^{2n+1} + y^{2n+1}$ is divisible by $x + y$

Statement - 2: If $n \in N$, $n^3 + 2n$ is divisible by 6

Which of the above statement is true:

 - only 1
 - only 2
 - both 1 & 2
 - neither 1 nor 2
1. $49^n + 16n - 1$ is divisible by $A(n \in N)$

2. $3^{2n} + 7$ is divisible by $B(n \in N)$

3. $4^n - 3n - 1$ is divisible by $C(n \in N)$

4. $3^{3n} - 26n - 1$ is divisible by $D(n \in N)$

then the increasing order of A, B, C, D is

 - A, B, C, D
 - C, B, A, D
 - B, C, A, D
 - D, A, C, B
- Statement - 1:** For all $n \in N$ $x^n - y^n$ is divisible by $x - y$

Statement - 2: $x^n + y^n$ is divisible by $x + y$ if n is even natural number

Which of the above statement is true :

 - only 1
 - only 2
 - both 1 & 2
 - neither 1 nor 2
- Assertion (A):** For all +ve, integral values of 'n', $3^{2n} + 7$ is divisible by 8

Reason (R): G.C.F. of 16 and 88 is 8

 - A true, R true and R is the correct explanation of A
 - A true, R true and R is not a correct explanation of A
 - A true, R false
 - A false, R true.
- Assertion (A):** $n \in N$ product of $n(n+1)(n+2)$ is divisible by 6

Reason (R): Product of 3 consecutive +ve integers is divisible by 3!

 - A true, R true and R is the correct explanation of A
 - A true, R true and R is not a correct explanation of A
 - A true, R false
 - A false, R true.
- Statement - 1:** For every natural number $n \geq 2$,

$$n \geq 2, \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > \sqrt{n}.$$

Statement - 2 :

For every natural number $n \geq 2$,

$$\sqrt{n(n+1)} < n+1.$$

[AIE- 2008]

- 1) Statement - 1 is true, Statement - 2 is true; Statement - 2 is not a correct explanation for Statement-1

2) Statement - 1 is true, Statement - 2 is false

3) Statement - 1 is false, Statement - 2 is true

4) Statement - 1 is true, Statement - 2 is true ; Statement - 2 is a correct explanation for Statement-1
- Assertion (A):**

$$n(n^2 - 1^2)(n^2 - 2^2) \dots (n^2 - r^2), n > r, n \in N$$

is divisible by $(2r+1)!$

Reason (R): Product of 'r' consecutive natural numbers is divisible by $r!$.

 - Both A and R are true and R is the correct Explanation of A
 - Both A and R are true but R is not the correct Explanation of A
 - A is true and R is false
 - A is false and R is true
- Statement-1:** The digit in the unit place of $183! + 3^{183}$ is 7.

Statement-2: $183!$ have unit place 0 and 3^{4k+3} ends with 7.

 - Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1.
 - Statement-1 is true, Statement-2 is false.
 - Statement-1 is false, Statement-2 is true.
 - Statement-1 is true, Statement-2 is true, Statement-2 is correct explanation for Statement-1.
- Statement-1:** If $A = (300)^{600}$, $B = 600!$, $C = (200)^{600}$, then $A > B > C$.

Statement-2: $\left(\frac{n}{2}\right)^n > n! > \left(\frac{n}{3}\right)^n$ for $n > 6$.

 - Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1
 - Statement-1 is true, Statement-2 is false.
 - Statement-1 is false, Statement-2 is true.
 - Statement-1 is true, Statement-2 is true, Statement-2 is correct explanation for Statement-1.

10. **Statement-1:** Let

$$P(x) = (x+1)(x+2)(x+3)\dots(x+11),$$

then coefficient of $x^{10} = 64$.

Statement-2: Coefficient of x^{n-1} in the product of

$$(x+1)(x+2)\dots(x+n) \text{ is } \frac{n(n+1)}{2}.$$

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

2) Statement-1 is true, Statement-2 is false.

3) Statement-1 is false, Statement-2 is true.

4) Statement-1 is true, Statement-2 is true, Statement-2 is correct explanation for Statement-1.

11. **Statement - 1 :- The sum of the series**

$$10. \quad 1 + (1+2+4) + (4+6+9) + (9+12+16) + \dots + (361+380+400) \text{ is } 8000$$

$$\text{Statement - 2 :- } \sum_{k=1}^n [k^3 - (k-1)^3] = n^3,$$

for any natural number 'n'

1) Statement-1 is true, Statement-2 is true, Statement-2 is not a correct explanation for Statement-1

2) Statement-1 is true, Statement-2 is false

3) Statement-1 is false, Statement-2 is true

4) Statement-1 is true, Statement-2 is true, Statement-2 is a correct explanation for Statement-1

12. **Statement - 1 :** For each natural number n,

$$(n+1)^7 - n^7 - 1 \text{ is divisible by } 7.$$

Statement - 2 : For each natural number n,

$$n^7 - n \text{ is divisible by } 7. \quad (\text{AIEEE-2011})$$

1) Statement - 1 is false, Statement-2 is true

2) Statement-1 is true, Statement - 2 is true & Statement - 2 is correct explanation of Statement-1

3) Statement-1 is true, Statement - 2 is true & Statement - 2 is not a correct explanation of Statement-1

4) Statement-1 is true, Statement-2 is false.

LEVEL - IV-KEY

- 01) 1 02) 3 03) 1 04) 1 05) 1 06) 1
07) 1 08) 4 09) 4 10) 3 11) 4 12) 2

LEVEL - IV-HINTS

- Put $n = 1$ and verify the options.
- Put $n = 1, 2, 2, 2$, $B = 4$, $C = 9$, $A = 64$, $D = 666$
- Put $n = 1, 2$ and verify the options.
- Put $n = 1, 2$ and take G.C.F
- Product of 3 successive integers is divisible by $3!$.
- By verification
- By observation
- For every $n! \forall n \geq 5$ ends with 0 and 3^{4n} ends with 1.
 $\therefore 3^{4n+3} = 3^{4n} \cdot 3^3$ where unit place is 7.

$$9. \quad \text{Put } n = 600 \Rightarrow \left(\frac{600}{2}\right)^{600} > 600! > (200)^{600} \\ \Rightarrow A > B > C$$

$$10. \quad (x+1)(x+2)\dots(x+n) \\ x^n + (1+2+\dots+n)x^{n-1} + \\ [(1 \cdot 2 + 1 \cdot 3 + \dots + 1 \cdot n) + \\ (2 \times 3 + \dots + 2 \times n) + \dots + (n+1) \times n]x^{n-2} \\ + \dots + (1 \times 2 \times \dots \times n) = \frac{n(n+1)}{2} \Rightarrow \text{coefficient of} \\ x^{10} = x^{11-1} = \frac{11(12)}{2} = 66 \neq 64$$

$$11. \quad \sum_{k=1}^n [k^3 - (k-1)^3] = \\ (1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots + \\ (n^3 - (n-1)^3) = n^3 \\ 1 + (1+2+4) + (4+6+9) + (9+12+16) + \dots + \\ (361+380+400) = \\ (1^3 - 0^3) + (2^3 - 1^3) + (3^3 - 2^3) + \dots + \\ (20^3 - 19^3) = 20^3 = 8000$$

$$12. \quad \text{For } n = k, P(k) = k^7 - k \\ \text{Let } P(k) \text{ is divisible by } 7. \\ k^7 - k = 7\lambda \text{ for some } \lambda \in N \dots (1) \text{ For } \\ n = K+1, P(k+1) = (k+1)^7 - (K+1) \\ = ({}^7C_0 k^7 + {}^7C_1 k^6 + {}^7C_2 k^5 + \dots + {}^7C_6 k + {}^7C_7) \\ - (K+1) = (K^7 - k) + 7(k^6 + 3k^5 + \dots + k) \\ = 7\lambda + 7(k^6 + 3k^5 + \dots + k)$$